SC452 — Processes on Networks

Course Project — Phase 1: Title and Outline

GROUP NUMBER

Group 5

GROUP MEMBERS

Name	Roll Number
Tirth Gandhi	202301016
Prince Chovatiya	202301067
Nakul Patel	202301261

PROJECT TITLE

Does Aggregation Lie? Epidemic Thresholds on Temporal vs. Static Contact Networks

Project Outline

Problem Statement

Epidemic models usually collapse a time-varying contact network into one static graph and estimate the SIS epidemic threshold from its largest eigenvalue ($\tau_c \approx 1/\lambda_1$). Real contacts are bursty and time-stamped, so we investigate how much this static approximation shifts the true threshold and outbreak size.

Course Concepts

Network structure (adjacency spectrum, degree distribution, k-cores, centrality); dynamical systems (fixed points and stability of SIS mean-field equations); dynamical processes on networks (SIS/SIR spreading, NIMFA mean-field approximation).

Methodology

We use real time-stamped datasets (SNAP email-Eu-core-temporal, CollegeMsg, SocioPatterns face-to-face contacts), building both a static aggregated graph and a sliding time-window sequence for each. We then run stochastic SIS/SIR simulations and NIMFA on both, using timestamp-shuffled Erdos–Rényi and Barabási–Albert graphs as a null baseline.

Expected Outcomes

Predicted vs. simulated thresholds for static and temporal representations, an estimate of how much error comes from aggregation versus burstiness, and a comparison of degree-based, k-core-based, and time-aware immunization strategies.